

Infrastructure Resource Report

Grand Vision

v2.0 · Richard Uzor — May 2026 · Devstrike Digital Limited

01 — PHILOSOPHY

Infrastructure spending follows the same trust-first principle as the product. Start on free tiers. Upgrade when metrics demand it — not when assumptions suggest it. Every upgrade trigger is explicitly defined.

02 — FULL RESOURCE STACK

Resource	Provider	Phase 1	Phase 2	Phase 3
Frontend	Vercel	Hobby (free)	Pro ₦32k/mo	Pro Team ₦80k/mo
API server	Railway	Starter (free)	₦16k-₦25k/mo	₦40k-₦80k/mo
Database				

Resource	Provider	Phase 1	Phase 2	Phase 3
	Railway PostgreSQL	Shared (free)	Dedicated ₦16k/mo	Dedicated + replica ₦40k/mo
Cache	Upstash Redis	Not required	Free tier	Pay-as-you-go ₦16k/mo
Job worker	Railway	N/A	₦8k-₦16k/mo	₦16k-₦24k/mo
WebSocket	Ably	N/A	N/A	₦32k-₦64k/mo
Object storage	Cloudflare R2	₦2k-₦4k/mo	₦8k-₦16k/mo	₦24k-₦48k/mo
CDN	Cloudflare	Free	Free	Pro ₦32k/mo
Email	Resend	Free	₦32k/mo	₦64k-₦96k/mo
Push notifications	FCM	Free	Free	Free
Error tracking	Sentry	Free	Free	Team ₦42k/mo
Analytics	Plausible	₦14k/mo	₦14k/mo	Business ₦30k/mo
AI embeddings	OpenAI API	N/A	N/A	₦32k-₦64k/mo
TOTAL		~₦20k/mo	~₦130k/mo	~₦450k-₦700k/mo

03 — UPGRADE TRIGGER SUMMARY

Resource	Upgrade Trigger
Vercel Hobby → Pro	Monthly bandwidth > 100GB, or team > 1 developer deploying
Railway free → paid	API response P95 > 400ms, or free credit exhausted
PostgreSQL shared → dedicated	Row count > 500,000, or daily writes > 50,000
Resend free → paid	Monthly emails > 2,400 (80% of free limit)

Resource	Upgrade Trigger
Upstash free → paid	Daily cache commands > 8,000
Cloudflare free → Pro	Needing custom WAF rules, or DDoS attacks exceed free mitigation
Sentry free → Team	Monthly errors > 5,000, or need performance monitoring
Add WebSocket server	Launching fellowship rooms in Phase 3
Add read replica	Read:write ratio > 10:1, or GET query P95 > 100ms

04 — DISASTER RECOVERY POLICY

Item	Policy
Database backups	Daily automated via Railway. 30-day retention from Phase 2.
RTO (Recovery Time Objective)	Phase 1-2: 4 hours. Phase 3: 1 hour.
RPO (Recovery Point Objective)	Phase 1-2: 24 hours. Phase 3: 1 hour (WAL streaming to replica).
Runbook	/docs/ops/runbook.md in GitHub repo. Reviewed quarterly.
Incident response	Sentry alert → team WhatsApp → triage 15 min → status page 30 min → post-mortem 48 hrs.